Understanding Infection-Associated Chronic Illness

A Public Guide to Long COVID, Post-Exertional Malaise, and the Science Behind the Diagnosis

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Chapter 1

Understanding Infection-Associated Chronic Illness

1.0.1 A Public Guide to Long COVID, Post-Exertional Malaise, and the Science Behind the Diagnosis

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First edition.

1.1 Who this book is for

You, or someone you love, has been told “your tests are normal” while still living with fatigue, brain fog, or exercise intolerance that never resolved after an infection. This book is for that reader: it does not assume a medical or research background, and it does not promise a cure. It exists to give you the same honest map researchers themselves are working from – what is established, what is a well-supported hypothesis still being tested, and what is genuinely contested – so you can read a headline, a new study, or your own chart with better judgment.

1.2 How this book is organized

Chapter 1 defines infection-associated chronic illness (IACI) itself – the umbrella term the NIH and the National Academies now use for the family of chronic conditions that follow an infection, and why that framing matters for how research and care get organized. Chapter 2 takes long COVID / post-COVID-19 condition as the case study: its three competing case definitions (WHO, RECOVER, NASEM), the

real uncertainty in its prevalence estimates, and why a number from one study is not always a claim about the same population as a number from another. Chapter 3 maps the leading mechanistic hypotheses – viral persistence, immune dysregulation, latent-virus reactivation, microclotting, mitochondrial dysfunction, dysautonomia, gut dysbiosis – honestly labeled by their actual evidence state, not by which sounds most convincing. Chapter 4 covers post-exertional malaise (PEM) and pacing, the symptom and the self-management practice that connects long COVID to myalgic encephalomyelitis (ME/CFS) and that dominates how patients actually have to live day to day. Chapter 5 teaches you to read the primary research yourself: what a case definition controls for, what a prevalence estimate's confidence interval is telling you, and how to tell a replicated finding from a single group's still-unconfirmed result.

1.3 A note on honesty

Nothing in this book is settled the way, say, germ theory is settled. Infection-associated chronic illness is an active, rapidly moving research field, and this book says so plainly wherever a claim is a hypothesis rather than an established finding, and cites the primary study behind every specific number or mechanism it names. This book is not medical advice and does not replace a clinician who knows your case; it exists so that the conversation you have with that clinician, or with a study you read on your own, starts from an accurate picture of what is and is not yet known.

Every chapter cites its sources. A bracketed marker like (Dehlia and Guthridge 2024) after a sentence points to a specific published study; the full citation – authors, journal, year, and a link you can look up yourself – is collected in the References section at the end of the book. You don't need to interrupt your reading to check these, but they are there whenever you want to see exactly where a number or claim came from.

Chapter 2

Chapter 1: What Infection-Associated Chronic Illness Is

2.1 An umbrella term for a pattern that keeps repeating

Let's start with a fact that researchers have noticed over the past few years: something happens after infection, and it keeps happening the same way, regardless of which infection triggers it.

When people survive certain infections — SARS-CoV-2 (COVID-19), the Ebola virus, dengue fever, and the bacterium that causes Lyme disease — a fraction of them experience chronic symptoms that persist for months or years. These symptoms look similar across different infections: profound fatigue that rest doesn't fix, a difficulty with thinking that people call "brain fog," and something called post-exertional malaise — a worsening of symptoms that comes a day or two after physical or cognitive exertion and can last for weeks (Choutka et al. 2022).

The pattern is consistent enough that in 2024, the National Academies of Sciences, Engineering, and Medicine — a research body that advises the U.S. government — decided to group these conditions together under one umbrella term: **infection-associated chronic illness**, or IACI for short (National Academies of Sciences, Engineering, and Medicine 2024b). The idea behind the umbrella is practical: if the pattern is the same across infections, maybe testing research across different illnesses can teach us something faster than studying each one alone.

This chapter explains what IACI is, which specific conditions it includes, and what researchers actually know versus what they're still trying to figure out.

2.2 Which conditions are in the umbrella

Infection-associated chronic illness includes six named conditions:

- **Long COVID** (also called post-COVID-19 condition, PASC, or post-acute sequelae of SARS-CoV-2 infection): chronic symptoms that appear after SARS-CoV-2 infection and persist for at least three months (National Academies of Sciences, Engineering, and Medicine 2024a).

- **Myalgic encephalomyelitis / chronic fatigue syndrome** (ME/CFS): a pre-existing condition (decades older than COVID) characterized by fatigue that rest does not fix, post-exertional malaise, and cognitive impairment (Institute of Medicine Committee on the Diagnostic Criteria for Myalgic Encephalomyelitis/Chronic Fatigue Syndrome 2015).
- **Post-treatment Lyme disease syndrome** (PTLDS): persistent pain, fatigue, or cognitive impairment that lasts after antibiotic treatment for *Borrelia burgdorferi* infection (the bacterium that causes Lyme disease) (Aucott et al. 2013).
- **Post-Ebola syndrome**: chronic symptoms including fatigue, headache, muscle pain, and memory loss that persist after Ebola infection (PREVAIL III Study Group, Sneller, and et al. 2019).
- **Post-dengue fatigue syndrome**: significant fatigue that persists after hospitalization for dengue fever (Seet, Quek, and Lim 2007).
- **Post-SARS chronic illness**: chronic fatigue and disability that persisted for years in survivors of the 2003 SARS-CoV-1 outbreak (Moldofsky and Patcai 2011; Lam and et al. 2009).

These six are not the whole list of post-infection syndromes, but they are the ones that IACI groups together because they show similar patterns and have attracted enough research interest to be measurable (National Academies of Sciences, Engineering, and Medicine 2024b).

2.3 Why this grouping matters

You might ask: why lump them together instead of treating each one separately?

The practical answer is a research problem. Individually, post-treatment Lyme disease syndrome or post-Ebola syndrome are rare enough and geographically scattered enough that it is nearly impossible to gather the number of patients needed to run a rigorous clinical trial or conduct the genetic and immunological studies that would reveal what's happening in the body. A single researcher in a single hospital might see a handful of cases a year.

But long COVID arrived suddenly, infecting hundreds of millions of people worldwide. A meaningful fraction of those people still have symptoms years later — a self-report tracking survey in the United States alone has put current prevalence at roughly 6-7% of all adults (Ford et al. 2023), and a global modeling effort estimated well over 100 million people worldwide experienced at least one persistent symptom cluster in 2020-2021 alone (Wulf Hanson et al. 2022). That scale created funding and research infrastructure that smaller post-infection illnesses had never had access to, including large, federally coordinated cohort studies like the NIH's RECOVER initiative, which enrolled nearly 10,000 participants to build a shared research definition of long COVID (Thaweethai et al. 2023).

The idea behind IACI is: test the hypotheses on long COVID, where you have millions of potential participants and large-scale research funding. Then take what you learn and ask whether it applies to the smaller, under-resourced conditions. If the mechanism that drives post-exertional malaise in long COVID is the same one driving it in ME/CFS, that finding helps both populations. If mechanisms differ, that difference is also useful to know.

As the National Academies put it, the IACI framing is a **strategy for improving research design**, not a claim that these six conditions are one disease (National Academies of Sciences, Engineering, and Medicine 2024b). Whether they turn out to share one mechanism, several mechanisms, or are etiologically distinct despite their symptom overlap — that is exactly the question the research agenda exists to answer.

2.4 The class symptom cluster

Despite their different infectious triggers, these conditions share overlapping core symptoms. The three that show up most consistently across all named IACI conditions are:

Fatigue that rest does not fix. This is not ordinary tiredness. It is not the fatigue after exertion that resolves with sleep. It is a baseline level of exhaustion that persists even with adequate rest, and that worsens with activity in a way that simple deconditioning does not explain (Choutka et al. 2022).

Cognitive impairment. People commonly describe this as "brain fog": difficulty with concentration, memory problems, or a general slowness in thinking that was not there before the infection. In some people it is mild; in others it is severe enough to affect work or daily functioning (Choutka et al. 2022).

Post-exertional malaise (PEM). This is the symptom that links long COVID most directly to ME/CFS. Post-exertional malaise is not ordinary muscle soreness from exercise. It is a disproportionate worsening of symptoms — fatigue, pain, cognitive fog, or autonomic symptoms like heart palpitations — that comes on characteristically 12 to 72 hours after physical, cognitive, or emotional exertion and lasts for days to weeks (Institute of Medicine Committee on the Diagnostic Criteria for Myalgic Encephalomyelitis/Chronic Fatigue Syndrome 2015). A person might take a short walk and feel fine immediately afterward, then become profoundly fatigued 24 or 48 hours later, with the worsening persisting for a week or longer.

Other symptoms show up frequently but not universally: difficulty regulating body temperature and blood pressure (dysautonomia), gastrointestinal problems, sleep that does not feel restorative, and various forms of pain (headaches, muscle or joint pain, or nerve pain).

2.5 What is established, what is being tested, and what is genuinely unknown

This is the most important section of this chapter, because it is where the book separates what is actually known from what *sounds* known.

2.5.1 Established findings

These facts are supported by evidence that has been tested and replicated across multiple studies:

1. **People do experience persistent symptoms after these infections.** This is not debated. Millions of people with long COVID have been documented. Studies of ME/CFS cohorts exist across multiple continents. Post-SARS chronic illness was documented in survivors from the 2003 outbreak, with symptoms lasting years (Moldofsky and Patcai 2011; Lam and et al. 2009), and post-Ebola sequelae have been documented in a large longitudinal cohort in Liberia (PREVAIL III Study Group, Sneller, and et al. 2019).

2. **A substantial share of people infected experience these symptoms.** Prevalence varies by how you measure it. The U.S. CDC's self-report household survey (asking people "do you have long COVID now?") finds roughly 6-7% of all adults report current long COVID (Ford et al. 2023). Smaller, clinically validated cohorts report lower numbers, but exact figures depend on which definition of long COVID you use — a problem discussed in Chapter 2. The point: it is not a tiny fraction, but it is also not everyone who is infected.

3. **Symptom clusters overlap across the IACI conditions.** Fatigue, post-exertional malaise, and cognitive impairment show up across long COVID, ME/CFS, PTLDS, post-Ebola, and post-dengue cohorts. Different cohorts emphasize different combinations, but the pattern is consistent enough to justify grouping them (Choutka et al. 2022; National Academies of Sciences, Engineering, and Medicine 2024b).

4. **These are not simply deconditioning or psychological illness.** This is an important boundary to state plainly. The fatigue and cognitive impairment do not resolve through standard approaches to deconditioning (graded exercise that gradually increases activity). The post-exertional malaise response — a delayed and prolonged symptom flare — does not fit patterns of ordinary fatigue or typical exercise response, a distinction the 2015 U.S. Institute of Medicine report made central to its diagnostic criteria (Institute of Medicine Committee on the Diagnostic Criteria for Myalgic Encephalomyelitis/Chronic

Fatigue Syndrome 2015). People with these conditions often did not have anxiety, depression, or other psychiatric illness before infection, and psychiatric symptoms do not fully explain the pattern. (Psychiatric symptoms *can* occur as a secondary consequence of living with a chronic disabling illness, which is a separate point.)

2.5.2 Mechanistic hypotheses actively being tested

These are candidate explanations for what causes IACI symptoms. They have supporting evidence, but that evidence is not yet conclusive, and different research groups are actively testing them:

- **Viral persistence**: SARS-CoV-2 viral RNA has been detected in the blood of some patients, a signal that is strongest around the time of diagnosis and drops sharply — two- to three-fold — by two to three months out, when very few patients still test positive (Su et al. 2022). Whether the material still present at those later, rarer detections is actively driving symptoms, or is an inert remnant of the infection, is not yet settled. One randomized, placebo-controlled trial testing an antiviral drug (nirmatrelvir-ritonavir) on long COVID patients found that it did not significantly improve symptoms compared to placebo, which complicates the simplest version of the "persistence causes symptoms" story without ruling out persistence as a contributing factor in some patients (Sawano et al. 2025).
- **Immune dysregulation**: Studies of long COVID show signs of abnormal immune activation: altered T-cell and B-cell populations, an exaggerated antibody response, and lower cortisol levels (Klein et al. 2023). Whether these represent a cause of symptoms or a consequence of chronic illness is still being determined.
- **Latent virus reactivation**: Some long COVID patients show signs of reactivation of Epstein-Barr virus (a common virus that usually remains dormant after childhood infection) (Gold et al. 2021; Su et al. 2022). Whether EBV reactivation is driving symptoms or is a consequence of immune dysregulation is not yet established.
- **Autonomic nervous system dysfunction**: Many people with long COVID have symptoms of dysautonomia — difficulty regulating heart rate, blood pressure, and other autonomic functions. Postural orthostatic tachycardia syndrome (POTS), a specific form of dysautonomia, has been documented in a significant minority of long COVID patients evaluated in specialized clinics (Björnson et al. 2025). Whether this is a primary mechanism or a secondary consequence is still being studied.

- **Metabolic and mitochondrial dysfunction**: Some research suggests that cells in people with long COVID are not producing energy efficiently. One study found that in long COVID patients with post-exertional malaise, a standardized exercise challenge triggered measurably worse abnormalities in muscle metabolism the following day (Appelman et al. 2024). Whether this is a primary cause of symptoms or a consequence of another upstream problem is not yet known.
- **Microbiome dysbiosis**: Alterations in gut bacterial composition at the time of hospital admission have been shown to predict long COVID status six months later in a prospective cohort, with the dysbiosis correlating with symptoms including fatigue and poor memory (Liu et al. 2022). Whether the dysbiosis is causing the illness or is a consequence of it is unknown.

2.5.3 Genuinely contested questions

These are areas where different research groups have published conflicting results or where the evidence is genuinely ambiguous:

- **Microclots and blood clotting abnormalities**: One research group has reported finding anomalous amyloid-form fibrin clots in long COVID blood samples that resist normal breakdown (Kell, Laubscher, and Pretorius 2022). Another group has published a direct critique arguing these microclots are not unique to long COVID, appear in other inflammatory and aging states, and that the measurement methods used across the field lack standardization (Hunt et al. 2024). This remains an active scientific dispute without resolution.
- **Whether G-protein-coupled receptor autoantibodies drive dysautonomia**: Some studies using functional assays (tests that measure biological activity on living cells) report high levels of autoantibodies against certain cell-surface receptors in long COVID patients, with correlations to specific symptoms (Hofmann et al. 2025). A separate case-control study using a standardized binding assay (a test that measures antibody binding to antigens fixed on a plate) found no meaningful difference between patients and controls for any of eleven receptors tested (Hall et al. 2022). The discrepancy may reflect differences in methodology — a functional assay and a binding assay measure different properties — rather than disagreement about biology, but it remains unresolved.

2.5.4 What is genuinely not yet studied

And finally, some important questions simply have not been adequately addressed yet:

- **Whether the same mechanism drives symptoms in all people with IACI, or whether different mechanisms produce the same symptom clusters in different people.** Long COVID is almost certainly mechanistically heterogeneous — different people may become ill through different pathways. Testing this requires large studies with detailed biomarking and careful phenotyping, which are only now being built.
- **Why the same infection produces different outcomes in different people** — why some people recover fully while others develop chronic symptoms. Early infection characteristics predict risk only partly (Su et al. 2022). Most of the variation remains unexplained.
- **Which findings from long COVID mechanistic studies actually transfer to ME/CFS, PTLDS, or post-Ebola conditions.** Long COVID and ME/CFS show overlapping symptoms, but mechanistic findings from long COVID studies cannot be automatically assumed to apply to ME/CFS. Testing this requires cross-condition research that is only beginning (National Academies of Sciences, Engineering, and Medicine 2024b).

2.6 What this means for reading the rest of this book

The landscape of IACI is one where:

- **Facts exist and are often surprising.** Infection does produce chronic illness in a measurable fraction of people. The symptoms are real and disabling.
- **Mechanisms remain largely unknown.** Multiple candidate explanations exist, all with supporting evidence, none conclusively proven.
- **Numbers require context.** A prevalence figure, a mechanism study, or a treatment trial is only interpretable if you know which definition of the disease it used, which population it sampled, and what it is measuring. The same condition studied by different researchers sometimes produces dramatically different numbers — not because the researchers are wrong, but because they asked subtly different populations different questions.
- **Honesty about uncertainty is the mark of good science, not a bug.** When researchers say “we don’t know yet,” they are being precise about what the evidence supports. That precision is what allows the field to move forward.

The remaining chapters walk through long COVID’s case definitions and prevalence (Chapter 2), the leading mechanistic hypotheses and their evidence state (Chapter 3), post-exertional malaise and the practice of pacing (Chapter 4), and

how to read primary research yourself (Chapter 5). Each teaches you to ask the questions that separate established findings from active hypotheses, so that when you read a headline or talk with a clinician, you are working from the same map the researchers themselves are using.

Chapter 3

Chapter 2: Long COVID — Definitions and Why They Disagree

3.1 The definition problem, plainly stated

If you read three different reports about how common long COVID is, or what the typical symptoms are, you may see three different numbers. This is not because the researchers disagree about the facts — it's because they're measuring different populations using different definitions. This sounds like a technical detail. It's not. It changes what any single study can actually tell you.

Here is the central fact that shapes everything else in this chapter: long COVID is not one thing measured three different ways. It is three different case definitions, each built for a different purpose, producing three different populations (Soriano et al. 2022; Thaweethai et al. 2023; National Academies of Sciences, Engineering, and Medicine 2024a). A finding from one is not automatically a finding about the other two.

3.2 Three definitions, three purposes

3.2.1 The WHO clinical case definition (2021–2022)

In 2021 and 2022, the World Health Organization convened a Delphi consensus — a structured expert process where 265 participants (patients, clinicians, researchers, and WHO staff) evaluated 14 different domains across two rounds and converged on a shared definition (Soriano et al. 2022). The WHO's definition is built for clinical diagnosis. If you walk into a doctor's office saying you've had COVID and you're still sick, this is what that doctor would use to figure out whether you have long COVID:

You have probable or confirmed SARS-CoV-2 infection; your symptoms started usually within three months of that infection; the symptoms have lasted at least two months; and they cannot be explained by an alternative diagnosis (a thyroid problem, depression, heart disease — something else that your doctor can identify and might explain your symptoms instead). The symptoms that matter most are fatigue, shortness of breath, and cognitive problems like brain fog — but you may

have others. The symptoms generally impact your everyday functioning and may be new symptoms that appeared after you got better from the acute COVID-19 illness, or they may be symptoms that just never went away (Soriano et al. 2022).

This definition is designed for diagnosis in a clinic and for comparing populations across countries. It is a consensus artifact — what the field agreed to call the condition — not a claim about what causes it.

3.2.2 The RECOVER research definition (2023)

The NIH RECOVER initiative took a different approach. Rather than starting with expert consensus, they started with data. They recruited 9,764 people — 8,646 who had been infected with SARS-CoV-2 and 1,118 who had not — and asked both groups about 44 different symptoms at six months or more after infection. Then they used statistical methods to figure out which symptoms actually separated the infected group from the uninfected group. From that analysis, they ended up with 12 symptoms that had the most power to distinguish the two groups (Thaweethai et al. 2023):

Post-exertional malaise (getting worse after activity), fatigue, brain fog, dizziness, gastrointestinal symptoms, heart palpitations, changes in sexual desire or capacity, loss of or change in smell or taste, thirst, chronic cough, chest pain, and abnormal movements (Thaweethai et al. 2023).

The RECOVER definition is not a diagnosis — it is a research instrument, a weighted composite score that researchers use to make sure they're studying the same population across multiple studies. It exists to give researchers a consistent outcome variable, a number they can measure and track. In the most methodologically rigorous subgroup RECOVER studied — participants first infected on or after December 2021 and enrolled within 30 days of infection — 10% were classified as having long COVID by this score at six months (Thaweethai et al. 2023).

3.2.3 The NASEM definition (2024)

In June 2024, the National Academies of Sciences, Engineering, and Medicine issued a definition for use by health systems and health policy (National Academies of Sciences, Engineering, and Medicine 2024a). This definition is deliberately broader. According to NASEM, long COVID is an infection-associated chronic condition that appears after SARS-CoV-2 infection and is present for at least three months as a continuous, relapsing-and-remitting (coming and going), or progressive disease state affecting one or more organ systems. It can present as a single symptom or as multiple diagnosable conditions — interstitial lung disease, heart arrhythmias, POTS (postural orthostatic tachycardia syndrome), ME/CFS (myalgic encephalomyelitis/chronic fatigue syndrome), diabetes, autoimmune disorders,

and others.

The NASEM definition widens the frame from “a symptom cluster” to “an umbrella disease state.” It folds in entities like ME/CFS and POTS that have their own entire research histories and their own case-definition debates. This changes what any prevalence or mechanism study can be read to claim about “long COVID” as a whole.

3.3 Why these definitions matter

The three definitions differ in timing, scope, and evidence base. The WHO definition requires symptoms to appear within about three months of infection and last at least two months — roughly a five-month window, and it is a consensus about what clinicians agree to call the condition (Soriano et al. 2022). The RECOVER definition assesses symptoms at six months or later post-infection with no separate timing requirement, and it is data-derived, showing which symptoms actually distinguish infected from uninfected people (Thaweethai et al. 2023). The NASEM definition requires three months duration with no upper bound, and it explicitly allows for the condition to improve, worsen, or come and go over time, rather than requiring a fixed course (National Academies of Sciences, Engineering, and Medicine 2024a).

Because these three instruments select different populations, findings produced by one study cannot be pooled with findings from another without that study stating which definition it used. This is the same validity problem that has been documented for myalgic encephalomyelitis for years — different case definitions of ME/CFS (Fukuda, Canadian Consensus, and the 2015 Institute of Medicine criteria) select measurably different populations, and a finding true for one is not automatically true for the other (Institute of Medicine Committee on the Diagnostic Criteria for Myalgic Encephalomyelitis/Chronic Fatigue Syndrome 2015). Long COVID inherited this same structural problem.

3.4 Prevalence: why the numbers disagree, and what the real uncertainty is

When you see a statistic about how many people have long COVID, stop and ask: what population was it measuring, and using which definition?

3.4.1 The CDC Household Pulse Survey: 6–7% of US adults

The CDC periodically asks a representative sample of noninstitutionalized US adults “Have you ever been told you have long COVID?” and “Do you currently have long

COVID?" This is a self-report survey with no clinical confirmation behind the answer. In June 2022, about 7.5% of US adults said they currently had long COVID. By June 2023, that figure had dropped to 6%, and among adults who had ever had COVID-19, the figure fell from 18.9% to 11.0% over the same interval (Ford et al. 2023). A March 2024 wave put it at roughly 7% again — about 17 million adults (National Center for Health Statistics, Centers for Disease Control and Prevention 2024–3AD). This number tells you what people self-report, which is useful for understanding how common people perceive the condition to be, but it does not tell you how many people actually meet a verified clinical case definition. It is a tracking trend, not a clinical case count.

3.4.2 The Global Burden of Disease modeling: 144.7 million people, with massive uncertainty

An international collaboration looked at long COVID across ten cohort studies in ten countries and modeled global prevalence. They organized the condition around three symptom clusters (fatigue, cognitive problems, and respiratory symptoms) and estimated that 144.7 million people experienced at least one cluster in 2020–2021 (Wulf Hanson et al. 2022). But here is the critical number that most discussions skip: the 95% uncertainty interval was 54.8 million to 312.9 million. That is a near-6-fold spread. The wide interval reflects real uncertainty in the data — different studies used different methods, different populations, different definitions. The finding is not "144.7 million people have long COVID." The finding is "we don't actually know the number, but we know it is large, and the true figure could be anywhere from one-third to more than two times the middle estimate."

3.4.3 The meta-analysis: 45% of survivors symptomatic at four months

A 2023 meta-analysis pooled studies of people who had COVID-19 and looked at what fraction had unresolved symptoms at approximately four months after infection. The pooled answer was 45% — nearly half of survivors had at least one lingering symptom (O'Mahoney, Routen, Gillies, and et al. 2023). But this figure drew published criticism in the same journal for pooling across inconsistent symptom-duration thresholds and including many studies with no comparison group of uninfected people. Some of the included studies used a very loose threshold (any lingering symptom at any point), while others required more sustained symptom persistence. The 45% is real within the pool of studies that met the inclusion criteria, but it may be inflated compared to a stricter, more consistent definition.

3.4.4 What these numbers actually tell you

Read together, these three approaches give you different answers to different questions. The CDC survey asks "What do people think they have?" and the answer is "6–7% self-report it" (Ford et al. 2023). The global modeling asks "If we tried to quantify the scope of the problem across all countries," and the answer is "somewhere between 55 million and 313 million, with a central estimate of 145 million, but we're genuinely uncertain" (Wulf Hanson et al. 2022). The meta-analysis asks "What fraction of COVID survivors have some lingering symptom at some point in the first four months?" and the answer is "45%, but only if we pool across studies that used different thresholds" (O'Mahoney, Routen, Gillies, and et al. 2023). None of these is wrong, and none is directly comparable to the other two.

This is not a flaw in the researchers. This is the cost of the definitional heterogeneity. Until the field converges on a single agreed case definition that researchers use consistently — the way they did for, say, diabetes or heart disease — any claim about prevalence will come with this caveat: "depends on which definition, and the definitions are still evolving."

3.5 The symptom clusters: what people actually experience

Across the three definitional frameworks, certain symptom domains appear consistently. These clusters matter because they are what people who have long COVID actually live with. This is not the mechanism — we do not yet know why these symptoms occur — but this is the phenomenology, the actual experience (Davis et al. 2023).

3.5.1 Fatigue and post-exertional malaise

The symptom that appears in nearly every definition is fatigue that is not relieved by rest (Thaweethai et al. 2023). People with long COVID describe an exhaustion that improves with sleep or resting but returns quickly, not a tiredness that gets better with downtime the way ordinary fatigue does. For a substantial fraction of people, the fatigue is accompanied by post-exertional malaise: when you undertake a physical activity, a cognitive task, or sometimes even an emotional exertion (like stress or worry), your symptoms get worse, often in a disproportionate way and often delayed — not right away, but 12 to 72 hours later. The worsening can last days to weeks. This is distinct from ordinary muscle soreness (which goes away in a day or two) and distinct from deconditioning (where a task produces immediate fatigue proportional to the effort). It is one of the most disabling features of long COVID for many people.

3.5.2 Brain fog and cognitive dysfunction

Memory problems, difficulty concentrating, word-finding difficulties, and a general sense of mental cloudiness — described by patients as "brain fog" — appear across the WHO, RECOVER, and NASEM frameworks (Soriano et al. 2022; Thaweethai et al. 2023; National Academies of Sciences, Engineering, and Medicine 2024a). Some people describe it as feeling like their thinking moves at half speed or their attention is foggy. Others report significant memory loss or persistent difficulty retrieving information they know they know. The severity ranges from noticeable but manageable to profoundly disabling.

3.5.3 Respiratory symptoms

Shortness of breath on exertion that is not explained by deconditioning, and chronic cough, appear consistently across definitions and cohorts (Wulf Hanson et al. 2022; Thaweethai et al. 2023). Some people experience dyspnea (difficulty breathing) even at rest or with minimal activity. Others notice the breathlessness only on exertion but find it more severe than it was before COVID.

3.5.4 Cardiovascular and autonomic symptoms

Heart palpitations, a sensation of the heart racing or skipping, appear in the RECOVER research definition (Thaweethai et al. 2023). A subset of people develop or worsen postural orthostatic tachycardia syndrome (POTS), a condition where your heart rate increases excessively when you stand up, and where you may feel dizzy, light-headed, or like you might pass out on standing. A prospective study of highly symptomatic long COVID clinic patients found that 31% met formal POTS criteria on standardized testing, with a further 27% having orthostatic symptoms without meeting full diagnostic thresholds (Björnson et al. 2025). The broader category of autonomic symptoms — symptoms related to the nervous system that controls heart rate, blood pressure, digestion, and other automatic functions — appears in a majority of people with long COVID who have been carefully evaluated in that same study.

3.5.5 Gastrointestinal symptoms

Nausea, vomiting, diarrhea, constipation, and abdominal pain appear in both the RECOVER research definition and in clinical cohorts (Thaweethai et al. 2023; Davis et al. 2023). Some people develop new food intolerances or changes in appetite.

3.5.6 Sensory changes

Loss of or change in sense of smell or taste was common early in the pandemic and appears in the RECOVER definition (Thaweethai et al. 2023). Some people recover these senses; others report persistent dysfunction months or years after infection. In addition, some people report sensory changes like tingling, numbness, or pins-and-needles sensations (Davis et al. 2023).

3.6 Symptom clusters, not one symptom in everyone

No single symptom is present in all people with long COVID. Some people have prominent fatigue and cognitive problems but minimal respiratory symptoms. Others have severe autonomic dysfunction and respiratory issues but less cognitive involvement. Some have profound post-exertional malaise; others have fatigue without the characteristic delay and worsening that defines post-exertional malaise. The global modeling work organized the condition around three clusters specifically — fatigue, cognitive problems, and respiratory symptoms — because those three had the strongest evidence across multiple studies, but this is not an exhaustive or universal list (Wulf Hanson et al. 2022).

The NASEM definition acknowledges this explicitly: long COVID can present as a single symptom or as multiple diagnosable conditions (National Academies of Sciences, Engineering, and Medicine 2024a). This heterogeneity is real. It means that if you have long COVID, your specific symptoms may not match someone else's profile, and that does not mean one of you does not actually have the condition. You both do; you just experience different aspects of it.

3.7 What this means if you're reading this

If you or someone you care about has had COVID-19 and is still experiencing symptoms, you now know three important things:

First, if a study says "45% of people have this" or "long COVID affects 7% of adults," you know to ask which definition they used and what population they measured. The same honest answer is usually "different definitions give different answers, and that's because the definitions are measuring different things for different purposes."

Second, you know that fatigue, post-exertional malaise, cognitive problems, shortness of breath, heart symptoms, and autonomic symptoms are real and documented, not rare edge cases or people being exhausted from anxiety or depression (though anxiety and depression are common alongside long COVID and are their own separate concern).

Third, you know that there is no single symptom profile for long COVID. If your symptoms do not match what a friend with long COVID experiences, that does not mean yours are not real or not long COVID. The condition is not one thing — it is a heterogeneous cluster of post-viral conditions that share a trigger (SARS-CoV-2 infection) and overlapping symptom domains.

The mechanisms — why these symptoms occur, what is happening in the body to cause them — are the subject of the next chapter. Here, the takeaway is simpler: long COVID is real, the symptoms are well-documented, the definitions are still evolving to be more precise, and the prevalence numbers disagree because the definitions disagree, not because researchers cannot count.

Chapter 4

Chapter 3: What Researchers Think Is Happening

When you recover from an infection, your immune system's job is to clear the virus and then stand down — return to baseline, stop the inflammation, let your body rebuild. That usually works. Occasionally it doesn't. When someone develops infection-associated chronic illness, something goes wrong with that stand-down. The question researchers are asking now is: what?

This chapter explains three lines of investigation into what might be happening. None of these is proven fact; all three are active research questions with real evidence pointing toward them but without final answers. What matters for you is understanding what the evidence shows, what remains uncertain, and how researchers are thinking about the problem.

4.1 The Persistence Theory: The Virus Isn't Entirely Gone

The first theory is the simplest to understand: maybe the virus isn't actually gone. Maybe somewhere in your body — in tissues, in organs, in the gut — viral material persists for months or years after the acute infection ended. Your immune system can see it but can't quite clear it. The virus keeps triggering low-grade inflammation, and that ongoing stimulus is what produces the chronic symptoms.

4.1.1 What's actually been found

Researchers have detected viral RNA and viral proteins in tissue samples from people with long COVID months after infection. Using ultrasensitive detection methods (tools so precise they can identify single molecules), scientists found circulating viral spike antigen in blood samples from about a quarter of people studied at various points in the first 14 months after infection (Swank et al. 2022; Peluso, Swank, et al. 2024). In one large study of gut tissue biopsies (249 patients), viral nucleocapsid protein (part of the virus's structure) was found in about 30 percent of patients sampled, at a median of about a year and a half after infection (Hany et al. 2023).

In one particularly striking study, researchers took gut biopsies from the rectosigmoid colon of five long-COVID patients and found viral RNA in all five — and double-stranded RNA (which suggests active viral replication, not just leftover RNA) in three of them (Peluso, Ryder, et al. 2024). At autopsy, researchers have recovered actual replication-competent virus — virus that can still reproduce, confirmed by growing it in cell culture — from tissue in the first two weeks or so after symptom onset. Separately, and further out, they have detected persistent viral RNA (genetic material from the virus, which is not the same as showing it is still actively reproducing) in multiple tissues, including the brain, as late as 230 days — about seven and a half months — after the acute illness in one case (Stein et al. 2022).

This is established: viral or viral-related material can persist in tissue and sometimes in blood for months to years. The persistence is real.

4.1.2 What remains unsettled

But persistence is not the same as causation. Just because viral material is still there doesn't automatically mean it's what's driving your symptoms. To establish that, you would need to show that clearing the virus also clears the symptoms. That test has been partly done: in a randomized controlled trial published in 2025, researchers gave long-COVID patients either an antiviral drug (nirmatrelvir-ritonavir) or placebo for 28 days (Sawano et al. 2025). The antiviral successfully reduced viral markers and broke the virus's replication cycle. Yet patients taking the antiviral did not improve more than the placebo group did. Their symptoms stayed the same.

This is a key finding: it suggests that whatever role persisting virus plays, killing it alone is not the full answer. That doesn't mean persistence is irrelevant — it could mean the antiviral was the wrong drug, the dose was wrong, the timing was wrong, or the damage is already done and the virus is no longer the active driver even though it's still present. But the simple story "virus persists → virus drives symptoms → kill virus → symptoms resolve" did not hold in this trial.

4.1.3 The evidence state

Viral persistence is **established as a real phenomenon**. Whether it is the primary driver of chronic symptoms remains a **mechanistic hypothesis still being tested**. One antiviral trial found no benefit for symptom resolution, but this is a single trial with a single drug and does not close the question.

4.2 The Immune Dysregulation Theory: The Immune System Is Running Wrong

The second major theory proposes something more complicated: the immune system itself is dysregulated. Not fighting an enemy anymore, but running in a strange, persistent state of partial activation — cells exhausted and activated at the same time, antibodies attacking the body's own tissue, inflammatory signals going out when there's nothing acute to fight.

4.2.1 What's actually been found

The immune dysregulation story divides into several parts, each with different evidence strength.

T-cell exhaustion. Researchers have found that a type of immune cell called CD8+ T cells, which normally fight viral infections, shows signs of exhaustion in long-COVID patients — specifically, high levels of PD-1, a protein that marks an exhausted immune cell that has been fighting for too long without clearing the enemy. These exhausted T cells appear in long-COVID cohorts and have been replicated in a second independent cohort by the same research group (Aid et al. 2025). This is the most robust single finding in the immune dysregulation literature, though independent replication by different labs is still in progress.

Autoantibodies. Several studies have found that long-COVID patients have elevated levels of antibodies that attack the body's own cells (Notarte et al. 2024). In one striking experiment, researchers took antibodies from long-COVID patients and injected them into mice: antibodies from one group of patients produced pronounced, lasting pain sensitivity in the mice, while antibodies from another group of patients reduced the mice's activity level in a way that resembles fatigue (Chen et al. 2026). This is genuine causal evidence in an animal model — injecting these antibodies produces disease-like effects in the animals. However, causal evidence in mice is not the same as proof of causation in humans, and whether these autoantibodies are the primary driver of long COVID in people remains an inference from the mouse data.

Complement dysregulation. The complement system is part of your innate immune machinery — a cascade of proteins that mark invaders for destruction and kill them. In long-COVID patients, multiple steps of the complement cascade are persistently activated: the classical pathway, the alternative pathway, and the terminal pathway all show markers of dysregulation (Baillie et al. 2024). This finding has been replicated across multiple studies and is established as a real phenomenon. However, different patients show different patterns of pathway activation, and the clinical significance of which exact pathways are dysregulated

remains contested — no study has yet shown that targeting a specific pathway improves symptoms (Notarte et al. 2026).

EBV reactivation. Epstein-Barr virus is a common virus that infects most humans by adulthood and then lives silently in the body for life. In some long-COVID patients, EBV reactivates — comes back out of hibernation — as if the immune system lost the ability to keep it suppressed. Studies have found that evidence of EBV reactivation appears more frequently in long-COVID patients than in fully recovered controls (Gold et al. 2021), and specific patterns of EBV antibodies are associated with fatigue and cognitive dysfunction in long-COVID populations (Peluso et al. 2023). However, EBV reactivation also occurs in some people without long COVID, and a substantial minority of long-COVID patients show no evidence of EBV reactivation at all — meaning it is associated with symptoms but is neither necessary nor sufficient to cause them. The question of whether EBV reactivation is driving the symptoms or whether the chronic illness is simply creating an environment where EBV can reactivate (two very different stories) remains open.

The contested finding: microclots. One research group has reported that blood from long-COVID patients contains microscopic clots with an unusual structure — amyloid-form fibrin that doesn't break down normally — and that these microclots impair oxygen delivery (Kell, Laubscher, and Pretorius 2022). Another group published a direct critique arguing that microclots are not specific to long COVID, that the methods for detecting and counting them lack standardization, and that the inference from seeing microclots to microclots causing symptoms is not evidence-based (Hunt et al. 2024). The originators responded with a rebuttal (Kell, Khan, and Pretorius 2024). This is an **unresolved scientific dispute** — both sides are named and present arguments in the published literature, and no independent, large-scale, controlled human trial has settled the question. This line is one to watch, but it is not established fact.

4.2.2 The evidence state

Immune dysregulation broadly is **well-supported as an associated finding** — multiple types of immune dysfunction occur in long COVID. T-cell exhaustion is **replicated but not yet independently confirmed by different labs**. Autoantibodies are **associated with long COVID and have causal evidence in animal models**, though human causality remains an inference. EBV reactivation is **established as an association with specific symptom clusters** but causality remains **mechanistic hypothesis**. Complement dysregulation is **established as a real phenomenon** but **contested in terms of clinical significance and which pathway matters most**. The microclot theory is **explicitly contested** and does not yet have standardized measurement methods or independent large-scale replication.

4.3 The Dysautonomia Theory: The Autonomic Nervous System Is Misfiring

Your autonomic nervous system runs your automatic functions — heart rate, blood pressure, digestion, sweating, breathing — the things you don't consciously control. It has two main arms: the sympathetic system (gas pedal, speeds things up) and the parasympathetic system (brake, slows things down). In healthy people, these two arms stay balanced: they rev up in response to stress or exertion, then calm down when the stress ends.

In dysautonomia, this balance breaks. The most common dysautonomic condition in long COVID is postural orthostatic tachycardia syndrome, or POTS — a state where standing up produces an excessive heart-rate increase and can cause dizziness, brain fog, fatigue, and other symptoms. But dysautonomia is broader than POTS alone; it's a general dysregulation of the autonomic system.

4.3.1 What's actually been found

Autonomic dysfunction is very common in long COVID. In studies of symptomatic long-COVID patients presenting to specialty clinics, about 31 percent meet formal diagnostic criteria for POTS, and an additional 27 percent have orthostatic symptoms that don't quite meet the formal criteria (Björnson et al. 2025). In the general population of people who had COVID-19, the rate drops to about 1 percent — much lower, but still higher than the rate in vaccinated people (about 0.04 percent), so infection does increase the risk (Yong et al. 2023).

The proposed mechanisms are two:

G-protein-coupled receptor autoimmunity. Your cells talk to each other through receptors — protein machines on the cell surface that receive signals. G-protein-coupled receptors (GPCRs) control things like heart rate, blood pressure, and breathing. Several single-center studies have found that long-COVID patients with POTS have antibodies that attack these receptors (Wallukat et al. 2021). A larger study of 194 long-COVID patients found high rates of these antibodies and showed associations between specific antibodies and specific symptoms like dizziness and concentration difficulty (Hofmann et al. 2025). However, one large, standardized case-control study using commercially available assay methods found no significant difference in GPCR autoantibodies between POTS patients and healthy controls (Hall et al. 2022). One possible explanation for this discrepancy is a difference in how the antibodies are measured (functional bioassays, which test whether the antibodies activate receptors, versus binding assays, which just test

whether they stick to the receptor) rather than a true disagreement about biology — though that reconciliation is this book's own interpretation of the two studies' methods, not a conclusion either study draws itself. **The mechanism is plausible but technically unresolved** — the evidence for the antibodies exists, but their causal role is contested.

Small-fiber neuropathy. The small nerve fibers in your skin that control autonomic functions (vasoconstriction, sweating) and carry pain signals can be damaged. In post-COVID populations, skin biopsy studies consistently find elevated rates of small-fiber denervation — loss of these delicate nerve fibers at the skin surface. One study found small-fiber neuropathy in 94 percent of post-COVID patients sampled, compared to much lower rates in uninfected controls. Critically, however, small-fiber neuropathy was present at similarly high rates (79 percent) in a second cohort of patients who developed similar sensory and autonomic symptoms after mRNA vaccination rather than after infection — not a healthy control group but a comparably symptomatic one — which complicates simple infection-specific explanations (Donadio et al. 2024). Clinicians have proposed treating this small-fiber neuropathy with intravenous immunoglobulin (a blood product containing antibodies), reasoning that if autoimmune antibodies are damaging the nerve fibers, an immune-modulating treatment might help; case reports describe improvement in some patients who received it, but the treatment has not been tested in a controlled trial, and the evidence remains preliminary (Dalakas 2024). The elevation of small-fiber neuropathy in post-COVID populations is **established as a real finding**, but whether this neuropathy is causally responsible for the orthostatic symptoms or whether it is a parallel consequence of the infection remains **mechanistic hypothesis**.

Management evidence is sparse. The first purpose-built randomized trial of ivabradine (a drug that slows heart rate) for long-COVID POTS, called RECOVER-AUTONOMIC, presented its results at the American College of Cardiology's 2026 conference, ahead of any full journal publication of those results: the drug successfully reduced heart rate but did not improve symptoms or quality-of-life measures. A secondary finding was that patients who also received coordinated nonpharmacologic care (pacing, compression garments, increased fluid intake) reported more symptom improvement than those on medication alone. (The trial's published design and rationale — written before results were known — is documented in (Fudim et al. 2026); the results themselves have so far been reported only at the conference.) Because these results have only been presented at a conference and have not yet passed peer review in a full published paper, they should be read as promising rather than settled — but they suggest that addressing dysautonomia in long COVID may require more than targeting the heart-rate problem alone.

4.3.2 The evidence state

Dysautonomia prevalence in long COVID is **well-established**. The proposed mechanisms — GPCR autoimmunity and small-fiber neuropathy — are both **associated with dysautonomia but mechanistically contested**. Whether either one is the primary driver versus an epiphenomenon or a parallel consequence is **not yet established**. Management evidence is limited; one antiarrhythmic drug worked pharmacologically but not symptomatically, and nonpharmacologic support appeared more beneficial in preliminary data.

4.4 How These Three Theories Might Fit Together

These three major lines of investigation — persistence, immune dysregulation, dysautonomia — are presented separately here, but in reality they may not be separate phenomena. Persisting virus could trigger immune dysregulation. Immune dysregulation could cause the autoantibody production that leads to dysautonomia. Small-fiber neuropathy could amplify dysautonomic symptoms. The three could be serial: one causes the next. Or they could be parallel: the initial infection damages multiple body systems in different ways, producing independent but coincident problems.

The research to distinguish these possibilities is ongoing. What matters now is understanding that **no single mechanism is proven as the cause of long COVID**, that **long COVID is likely mechanistically different in different patients**, and that **multiple lines of investigation are pointing at real biological abnormalities** even if the causal stories aren't complete yet.

4.5 The Bottom Line on Mechanisms

Researchers have identified real, measurable abnormalities in long-COVID patients:

- Viral or viral-related material persists in tissue for months to years in a subset of patients.
- Multiple types of immune dysregulation occur: T-cell exhaustion, autoantibodies, complement activation, and EBV reactivation.
- Dysautonomia, particularly small-fiber neuropathy and dysregulated autonomic reflexes, occurs at elevated rates.

None of these has been conclusively proven to be the driving cause of all long-COVID symptoms. One antiviral trial did not produce symptom improvement despite successfully lowering viral markers. Immune dysfunction occurs in long COVID but also in recovered patients, sometimes at similar rates. Dysautonomic findings are reproducible but could be causes, consequences, or parallel effects of the infection.

This is not a failure of the research — it is honest science. The researchers studying these mechanisms are holding the line between "here is what we found" and "therefore this causes the disease." That line is the difference between good science and false confidence, and it is worth respecting.

What this means for living with these conditions is that **no single treatment is likely to work for everyone**, because the underlying drivers are probably not the same in everyone. It also means **treatment research is still early** — we are at the stage of identifying candidate mechanisms, not at the stage of having proven targets for therapeutic intervention. But it means the research is real, the findings are substantive, and the field is asking the right questions even if the final answers are not yet in.

Chapter 5

Chapter 4: Post-Exertional Malaise and Pacing

5.1 The cardinal feature you cannot see immediately

Post-exertional malaise — or PEM, as it is often abbreviated — is the symptom that most sharply distinguishes the experience of living with long COVID or ME/CFS from living with ordinary fatigue, deconditioning, or the aftermath of a flu. It is also the symptom most people encounter by accident, often by learning it the hard way.

Post-exertional malaise is a disproportionate worsening of your symptoms following exertion. The exertion can be physical — a walk, cleaning the kitchen, climbing stairs. It can be cognitive — concentrating on work, following a conversation, reading for an hour. It can be emotional — stress, conflict, or simply the effort of maintaining composure. The critical feature is not which kind of exertion triggers it, but what happens afterward.

In ordinary fatigue, if you take a walk, you might feel tired immediately. You sit down, rest an hour, and feel better. In deconditioning — when your aerobic capacity has declined from lack of activity — exertion might produce muscle soreness that peaks a day or two later and fades within a few days. Post-exertional malaise is different on two counts: the severity and the timing.

With post-exertional malaise, a modest exertion you may not have even noticed in the moment — a short walk, light housekeeping, thirty minutes at a desk — can produce a severe flare 12 to 72 hours later. That flare is not proportional to the effort. A person might undertake a task that would cost a healthy person almost nothing, and then spend the next week barely able to leave bed. The fatigue is profound. Cognitive symptoms worsen — thinking becomes cloudy, words don't come, concentrating is impossible. Pain, fever-like sensations, sore throat, and autonomic symptoms (dizziness, heart palpitations, temperature regulation gone haywire) may flare alongside the fatigue. The flare lasts days to weeks, not hours (Stussman et al. 2020). In long COVID specifically, more than half of patients report experiencing post-exertional malaise within the first few months after infection (An et al. 2024).

This distinction matters because it changes how you have to think about activity.

An ordinary approach to fatigue is "rest when you need it, push a bit when you can, and gradually build capacity." That approach is calibrated for conditions where the cost of activity is paid immediately and quickly. In post-exertional malaise, the cost is delayed, disproportionate, and lingers. The same approach can backfire catastrophically.

5.2 Why pushing through doesn't work — and can make things worse

Most of us were taught to push through mild fatigue. "No pain, no gain." "Rest is for the weak." "You'll feel better once you get moving." These maxims work fine when the condition is deconditioning, acute illness, or depression — often, they contain a grain of truth. When the condition is post-exertional malaise, they become actively harmful.

Here is what happens when someone with post-exertional malaise tries to "push through":

You feel tired this morning, but not unbearably so. You decide to do the things you need to do anyway: go to work, run the errands, tackle the backlog on your desk. You might feel fairly okay through the day. Maybe you feel a little worse than baseline, but not dramatically. You go to bed thinking you've had a productive day.

Twelve to 48 hours later, the flare hits. Your fatigue is now profound. Your brain is fog. You cannot work, cannot cook, may barely be able to get out of bed. This wasn't supposed to happen — you felt okay yesterday. But the exertion you undertook has triggered an inflammatory and metabolic response that unfolds over hours, peaks over a day or two, and can persist for a week or weeks.

If you then make the mistake that many people do — thinking "I need to keep pushing to recover from this," or "resting will make me weaker" — you may undertake more activity while in the flare. This triggers another cycle, on top of the one still resolving. The flares can stack. What might have been a one-week setback from a single overactivity now becomes a monthslong decline from repeated stacked crashes.

Research on post-exertional malaise in ME/CFS has documented this pattern repeatedly in patient accounts. Patients describe the experience as a "crash" — an abrupt and severe worsening that feels separable from ordinary tiredness (Stussman et al. 2020). Studies using an objective measure called two-day exercise testing — a maximal exercise test repeated a day later — have often found exercise capacity measurably lower on day two than on day one in people with post-exertional malaise, the opposite of what happens in healthy people or people with simple deconditioning (Keller, Pryor, and Giloteaux 2014; Snell et al. 2013). This particular finding is not settled, and it is worth being honest about that: a larger 2026 study

using the same two-day protocol found no such decline, so researchers are still working out how reliable this test is as a marker of post-exertional malaise, and it should not be treated as a proven biological signature of the condition (Mancini et al. 2026).

The mechanistic picture researchers are building involves dysregulation of the systems that normally support exertion. One line of direct evidence: when long-COVID patients with post-exertional malaise were put through a standardized two-day exercise challenge, their muscle tissue showed measurably worse abnormalities the day after exertion — lower capacity to produce energy aerobically, and a shift toward less-efficient metabolism — though this comes from a single study that has not yet been replicated at a larger scale (Appelman et al. 2024). Researchers have also proposed a broader hypothesis to explain why the flare is delayed and prolonged rather than immediate: that the hormonal stress response, centered on a hormone called cortisol, normally helps mobilize fuel for exertion and then provides an anti-inflammatory brake that ends the body's inflammatory response once the exertion stops, and that in post-exertional malaise this brake may be engaged less effectively or delayed, allowing immune activation — cytokine release, oxidative stress, microglial activation — to unfold over hours and persist for days. This account fits the known biology and the delayed, prolonged shape of the flare, but it remains a hypothesis under active study, not an established mechanism.

The consequence, if you understand it, is that ignoring post-exertional malaise and "just pushing through" is not grit or strength. It is directly counterproductive. You are not overcoming the condition; you are repeatedly triggering the very response that drives the symptom worse.

5.3 What pacing actually is

This is where pacing comes in. Pacing, in the context of post-exertional malaise, is not slow motion or laziness. It is not "rest when you can manage it." It is a deliberate management strategy built on understanding how post-exertional malaise works.

The core idea is this: instead of doing as much as you can and then crashing, you do less than you could, consistently, so you never trigger the crash. You stay within an energy envelope — an estimate of how much activity you can undertake without triggering post-exertional malaise — and you respect that envelope, even when you feel like you could do more.

Finding your energy envelope is individual work. Some people track it by time — "I can do 20 minutes of activity before symptoms worsen." Some track it by specific tasks — "I can walk to the mailbox but not walk a mile." Some track a combination: cognitive work for an hour, then physical rest; light physical tasks

for 30 minutes, then sitting down. The envelope is different for every person, and it can change from week to week depending on baseline symptom severity, stress, sleep, and other factors.

The discipline is staying within it, even when you feel better. If your envelope is 20 minutes of activity per day, and you feel okay at 15 minutes, the temptation is to keep going — you're doing well, why stop? But the point of pacing is not to maximize the good moments. It is to avoid the crashes that make the bad months. You stop at 15 or 18 minutes, rest, and live to do it again tomorrow, without a flare.

Some people call this "energy accounting" — you budget your energy like you budget money, spending enough to live your life but not so much that you go into debt. Others call it "activity pacing" or "symptom-contingent pacing," which names the same idea. The vocabulary varies. The core strategy is identical: intentional limitation to prevent involuntary incapacity.

5.4 The honest evidence state for pacing

Here is where this book has to be honest about what we actually know and what we still don't.

Pacing is widely recommended for post-exertional malaise and ME/CFS. It appears in clinical guidelines — the UK's National Institute for Health and Care Excellence, for instance, recommends pacing and specifically advises against fixed-schedule graded exercise programs for ME/CFS (National Institute for Health and Care Excellence 2021). Patients report that it is helpful, often essential, for managing their symptoms and preserving function. The mechanistic reasoning — that avoiding exertion-induced flares would prevent the cascade of immune dysregulation and metabolic worsening — is sound.

But the rigorous controlled evidence for pacing is limited. A 2024 systematic review searching for controlled studies of pacing found only five that fit — two randomized controlled trials, one uncontrolled trial, one case series, and one subanalysis of a larger trial — and pooling them found the largest effect for fatigue, coming closer to conventional statistical significance than the other outcomes but not reaching it, while physical function and pain showed smaller effects that the review's own authors described as trivial and that fell further from statistical significance (Sanal-Hayes et al. 2024). Most of the evidence comes from patient reports, small observational studies, and case series. This does not mean pacing is not effective — it means we have not yet done the kind of systematic research that would tell us, with confidence, how much it helps, for whom it helps, and whether some versions work better than others.

This gap is partly a problem of research funding and infrastructure: post-exertional malaise is less widely recognized outside ME/CFS communities, clinical trials are

expensive, and long COVID research funding only recently became available. It is partly a problem of measurement: defining a pacing protocol precisely enough to test it, tracking activity and symptoms systematically, and having enough statistical power to detect differences all present real methodological challenges.

What we do have are several pieces of evidence pointing in a consistent direction:

Patient experience: The most immediate evidence comes from people living with post-exertional malaise themselves. Consistent across decades of ME/CFS patient accounts and increasingly common in long COVID patient reports is the finding that activity pacing prevents crashes, while activity without pacing does not (Stussman et al. 2020; Jason and Dorri 2022). When patients violate their energy envelope, flares follow. When they respect it, they remain stable or improve slowly over time. This is not random — the pattern is robust enough to be taught in patient education and used in self-management programs. But it is observational, not controlled.

Mechanism: The physiological reasoning is sound. Post-exertional malaise appears to involve dysregulated immune and metabolic responses to exertion (Appelman et al. 2024). If you avoid the exertion, you avoid triggering those responses. This is not proof — many sound mechanistic ideas don't pan out when tested — but it is a reason to think pacing might work.

Comparative harm: The evidence against *not* pacing is clearer than the evidence for pacing specifically. Graded exercise therapy — the approach of steadily increasing activity regardless of symptom response — has been associated with symptom worsening in ME/CFS patients, a finding directly documented in a patient-reported-outcome survey in which more than half of respondents reported large negative responses to graded exercise therapy (Geraghty, Hann, and Kurtev 2017). A separate freedom-of-information survey of NHS ME/CFS specialist clinics found that none had systematic protocols for monitoring or recording treatment-related harm, and that no clinic reported a single case of harm despite acknowledging that many patients dropped out of treatment — suggesting harm may go undetected rather than not occur (McPhee et al. 2019). This suggests that an activity approach that ignores post-exertional malaise can cause harm. Pacing, which is designed specifically to avoid triggering post-exertional malaise, should theoretically avoid that harm. The evidence supports the intuition, though not yet at the level of a definitive trial.

Trial evidence: A large NIH-funded platform trial testing management approaches for long-COVID dysautonomia, called RECOVER-AUTONOMIC, presented results at the American College of Cardiology's 2026 conference, ahead of any full journal publication of those results: participants who received structured nonpharmacologic coordination (which includes activity pacing guidance) reported greater

symptom improvement than those on medication (a heart-rate-slowing drug) alone. (The trial's published design and rationale — written before results were known — is documented in (Fudim et al. 2026); the results themselves have so far been reported only at the conference, not yet in an indexed paper.) This is real trial evidence, though it is still preliminary and has not yet passed peer review, and the trial was designed for dysautonomia management, not specifically for post-exertional malaise. It points in a hopeful direction.

5.5 What this means for someone living with it

If you are living with post-exertional malaise, here is the honest summary:

You have every reason to believe that pacing — staying within an energy envelope and avoiding activity that triggers crashes — will help preserve your function and quality of life. The physiological reasoning makes sense. Decades of patient experience with ME/CFS and the emerging experience with long COVID suggest it works. Early trial evidence in related conditions points in that direction.

But you are also living with something that has not yet been studied with the rigor we would want. There are no randomized trials proving pacing works better than alternatives, no studies definitively telling you the exact formula for your energy envelope, no research giving you confidence that this specific pacing protocol is optimal for your particular situation.

This is not a reason not to pace. It is a reason to approach pacing thoughtfully: to track your own experience, to notice what activity patterns lead to flares and what patterns keep you stable, and to adjust your envelope based on what actually happens to your body, not on guessing. Your own data — what activity level maintained stability for you this week, where the crashes happened, what recovered you — is valuable evidence. It will not answer the bigger research questions. But it will tell you what works for you.

It is also a reason to advocate for better research. People with post-exertional malaise need randomized trials comparing activity approaches, mechanistic studies of how pacing affects the immune and metabolic dysregulation underlying the symptom, and long-term follow-up studies tracking whether pacing prevents progression or enables recovery. Those trials are overdue. Until they happen, the honest epistemic position is: pacing is recommended, sounds reasonable, seems to help based on patient experience and early evidence — but we have not yet proven it the way we would prove a drug or a new surgical technique.

You can live with that uncertainty and still make a reasoned decision to pace. Many people do, every day.

Chapter 6

Chapter 5: How to Read Research About Your Own Illness

Every week brings a new headline about long COVID, ME/CFS, or infection-associated illness: a new study, a new mechanism, sometimes a new treatment that sounds promising. It is easy to read these headlines and either feel false hope or despair, depending on the day. The skill this chapter teaches you is simpler than it sounds: you don't need a PhD to read the actual research. You need to know what to look for — what makes a finding strong or weak, reproducible or preliminary, something that applies to *your* illness or to a different population entirely. This chapter walks through the real examples from the research we've discussed, showing you exactly how to spot the difference.

6.1 Case definitions: why the same study might not be about your condition

The single biggest source of confusion in reading research about infection-associated illness is that different studies are often measuring *different populations*, even though they use the same condition name. This is not a bug in the research — it is an honest result of researchers building on each other's work over years — but it means you have to look at which definition a study used before you can know whether it is studying *your* condition or a related one.

Here is a concrete example. In Chapter 2, we saw that long COVID has three major case definitions. A study of long COVID patients selected by the WHO clinical definition (symptoms usually starting around three months after infection and lasting at least two months, no alternative diagnosis) (Soriano et al. 2022) is measuring a *different population* than a study selected by the RECOVER research definition (using a weighted score of twelve specific symptoms assessed at six months or more) (Thaweethai et al. 2023). Not better, not worse — genuinely different people. This matters because the mechanistic findings from one definition do not automatically transfer to the other.

When you read a study about long COVID, the first thing to check is: *which case definition did they use?* Look for a section called "Methods" or "Case Definition"

early in the paper. If a study does not clearly state which definition it used, that is a red flag. A study that says "we studied long COVID patients" without saying *how* they selected them is a claim without enough evidence to act on.

Here is what you might actually find:

The straightforward case: A study explicitly states "we used the 2024 NASEM definition of long COVID: a chronic condition present at least three months post-infection, affecting one or more organ systems, with these specific inclusion criteria." That tells you exactly who the study looked at. If that matches your own experience, the findings may be relevant to you. If the study focused on patients with diagnosed myalgic encephalomyelitis, and you do not have that diagnosis, the finding applies to a subset, not necessarily to you.

The comparison problem: You read two studies that both claim to study "long COVID prevalence." One reports 6% of all US adults have current long COVID (this is the CDC Household Pulse Survey, which is self-report with no formal case definition) (Ford et al. 2023). Another reports 45% of COVID survivors were symptomatic at four months (the O'Mahoney meta-analysis, which pooled studies with heterogeneous definitions) (O'Mahoney, Routen, Gillies, and et al. 2023). These two numbers are not contradictory — they are answering *different questions* using *different methods*. The first is "what do Americans say when asked directly?" The second is "of people who were sick at four months, how many had unresolved symptoms?" The same person could fall into one category and not the other, or neither. The O'Mahoney number is itself contested: other researchers, writing a critique letter in the same journal, recalculated the pooled prevalence directly from O'Mahoney's own supplementary data and got 11% (86,552 out of 735,006 people), not 45% — attributing most of the gap to how the studies were pooled and weighted (Yao et al. 2023). They also pointed out that some of the 194 pooled studies followed patients for less than about four months, and cited other estimates that landed much lower (the CDC put current long COVID under 20%; one other study put it at 5.3%-9.8%), with prevalence ranging from 3% to 91% across the individual studies that went into the pool. O'Mahoney and colleagues published a reply defending their original methods (O'Mahoney, Routen, Gillies, and Khunti 2023). Neither side has retracted anything; this is an open methodological dispute about how to pool studies that used very different definitions and follow-up windows, not a case of one side simply being wrong.

When you read prevalence numbers, ask yourself: *is this the same population as the last study I read?* If not, the numbers cannot be compared as though one is more "correct." They are measurements of different things.

6.2 Sample size, controls, and robustness

The next thing to look for when reading a study is *how many people* the researchers studied and *whether they had a comparison group*. Bigger studies are not always better, but they are often more trustworthy. A study of nine people can generate a real, important observation — but that observation is preliminary until somebody else repeats it with a larger group or more rigorous methods.

Here is an example with real consequences. In 2023, a team published a small study (n=63 people with long COVID) showing that circulating spike protein antigen was detected in 60% of patients with long COVID and 0% of recovered COVID patients (Swank et al. 2022). This was the first report of this finding and it got attention — it seemed like researchers had found a biological marker specific to long COVID.

But here is what happened next: larger studies with better methods tried to repeat that finding.

Later, a more rigorous, blinded study examined 425 people: 167 with long COVID, 148 who recovered fully from COVID, and 110 who never had COVID (the uninfected controls) (Mateu et al. 2026). Crucially, the researchers measured antigen in a *blinded* way — they did not know who had long COVID and who recovered when they ran the tests, so their expectations could not unconsciously influence the results. And they used a comparison group that had never been infected, to check whether the assay itself was giving false positives.

What did they find? Yes, antigen was detected in people with long COVID (31% at six to twelve months). But antigen was *also* detected in 20% of people who had fully recovered from COVID. By eighteen to twenty-four months, antigen had become rare in all groups. Most importantly: antigen was not associated with the *number* or *type* of symptoms. People with the same antigen levels had wildly different symptom burdens.

This directly contradicted the first study's claim that antigen was specific to long COVID.

What changed between the two studies? Three things: **sample size** (63 vs 425), **blinding** (the second study was blinded to clinical status, the first was not), and **controls** (the second study included uninfected people to test the assay itself). These three things are what make the larger study more trustworthy.

When you read a study, look for:

1. **How many people?** A study with n=10 is preliminary. A study with n=425 is much more robust. The numbers alone do not make a study true, but larger numbers mean a more reliable estimate.

2. **Did they have a control group?** Did they compare patients to uninfected people, or to people with the illness who did not have symptoms? A study that

only looks at sick people cannot tell whether something is specific to illness or present in everyone.

3. **Was the study blinded?** In a blinded study, the person measuring the outcome (the blood test, the imaging scan, the questionnaire score) does not know which group the person belongs to. This prevents expectations from subtly influencing how results are measured or interpreted. It is a simple protection against a very human bias.

4. **Was the study a single site or did it combine data from multiple sites?** Single-site studies are valuable for generating a hypothesis. When the same finding shows up at multiple sites, independently, that is much stronger evidence.

When you read that one study found something and a larger, blinded study contradicted it, the larger blinded study is more trustworthy — not because the small study was wrong, but because the bigger study had more rigorous methods.

6.3 Replication: the difference between preliminary and established

Here is a pattern you will see repeatedly in research on infection-associated illness: a team reports an interesting finding, sometimes with real clinical implications. Then either (1) other teams independently confirm the finding, or (2) larger, more rigorous studies fail to confirm it, or (3) the finding sits alone, unreplicated, sometimes because nobody is interested in confirming it and sometimes because they tried and did not find the same thing.

Small studies are not inherently unreliable — they are how research starts, by spotting a pattern worth investigating further. But a preliminary finding, no matter how interesting, stays preliminary until somebody else can repeat it. And "somebody else" matters. If the same research team repeats their own finding in a new cohort, that is valuable, but it is not as strong as when an independent team, using their own cohort, finds the same thing.

Here is a real example. Researchers from one group have reported that autoantibodies (the immune system's own antibodies, attacking the body's own proteins) directed at G-protein-coupled receptors are associated with long COVID dysautonomia. Using a custom functional test, they found these autoantibodies in nearly all of the 31 recovered COVID-19 patients they tested (Wallukat et al. 2021), and a larger follow-up study by the same group, in 194 patients, found similar high rates along with correlations to specific symptoms like dizziness and difficulty concentrating (Hofmann et al. 2025). This is a compelling hypothesis.

But here is what happened when researchers from a different institution, using a standardized commercial test instead of the first team's custom functional assay, tried to detect these autoantibodies in a broader group of POTS patients (POTS from all causes, not just long COVID) compared against healthy controls: they found no significant difference between the two groups. Their best single-receptor discrimination was barely better than a coin flip (Hall et al. 2022).

This does not prove the first group was wrong. It suggests — strongly — that whatever they are detecting with their custom assays, it is not a simple, robust marker that shows up the same way every time, in every lab, using standard methods. The mechanism may be real. The marker may be real. But the "nearly all patients positive" claim only holds up in the group that developed the test.

When you read about a new mechanism or marker, ask: *Has anybody else confirmed this?* If it is from a single team, is it preliminary or established? If multiple teams have confirmed it using different methods, that is much stronger. If teams have tried and failed to confirm it, that is not evidence against the mechanism, but it is evidence against treating this particular marker as settled.

6.4 Confidence intervals and what uncertainty actually means

When you read a study that says something like "6% of US adults currently have long COVID" (Ford et al. 2023), that number is often paired with something called a confidence interval: a range around the headline figure. This range is not a guess about error. It is a precise statement about what the study's uncertainty actually is.

What does "95% confidence interval" mean? It means: if you repeated this survey or study many times, 95% of the time the true percentage would fall within this range. It does not mean "we are 95% sure the true answer is in this specific range." It means "this method, applied this many times, produces a range that contains the true answer 95% of the time." The distinction sounds technical, but it is important: it is about the method's reliability, not your personal certainty.

The width of the confidence interval tells you something important: how much uncertainty is baked into the estimate. This is a real, published estimate: a meta-analysis pooling 13 studies found that 51% of long COVID patients meet ME/CFS criteria (95% CI: 42%-60%) (Dehlia and Guthridge 2024). That wide interval (an 18-percentage-point spread) tells you the estimate has substantial uncertainty. This is not a weakness in the study — it is an honest statement that the true percentage could reasonably be anywhere in that range. A narrow interval, like 51% (95% CI: 49%-53%), tells you the estimate is more precise.

Here is a striking example from the research: the Global Burden of Disease

study estimated that 144.7 million people worldwide had long COVID in 2020-2021 (Wulf Hanson et al. 2022). The confidence interval? 54.8 million to 312.9 million. That is a six-fold spread — the low estimate is a bit over a third of the headline number, and the high estimate is more than double it. Does that mean the study is worthless? No. It means the researchers were honest about how much uncertainty there is when you try to model a condition globally, across different countries, with different definitions and different data quality. The wide interval is important information: it tells you this number is a rough estimate, not a precise count.

When you read a number with a confidence interval, look at the width of the range. A narrow range is more precise; a wide range is more uncertain. Either can be valuable information, as long as you know which one you are looking at.

6.5 Single studies versus replicated findings

The hardest skill in reading research is knowing when *one study's result* is a genuine discovery and when it is a preliminary observation that might not hold up. You cannot look at a single study and know this with certainty. But you can look for signs of robustness.

A classic example is the ivabradine trial we discussed in Chapter 4. Researchers tested a drug called ivabradine against placebo in 181 people with long COVID dysautonomia (RECOVER Initiative, National Institutes of Health 2026). It was actually a 2-by-2 trial: it also crossed each drug arm with either coordinated non-medication care (structured guidance on hydration, compression, exercise pacing, and the like) or usual care, so every patient landed in one of four combinations. On the drug alone, it worked exactly as expected: ivabradine slowed the heart rate. But by itself, that did not translate into patients feeling better — their orthostatic-intolerance symptom scores did not improve on ivabradine alone versus placebo alone. The combination arm told a different story: patients who got ivabradine *together with* the coordinated non-medication care showed a significantly greater improvement in symptoms than placebo (the interaction between drug and care approach was statistically significant, p=0.004). Heart rate going down did not, by itself, fix the symptom — but heart rate control paired with structured non-drug care did better than either alone.

This is not a study failure. It is a study that asks an important question — does fixing the heart rate fix the problem? — and gets a more interesting answer than a flat yes or no: not by itself, but combined with coordinated care, patients did better. That is a good illustration of this chapter's own point: a single variable in isolation (the drug alone) can look like a negative result, while the same variable combined with another intervention shows a real effect. (As of this writing, these results have only been presented at a cardiology conference and have not yet appeared in

a peer-reviewed journal — worth knowing, because conference results sometimes change during the peer-review process.)

But here is the limitation: it was *one trial*, with *one drug*, on *one primary outcome*, and the combined-care result came out of a single platform trial's factorial design, not a dedicated replication. Other drugs might work differently. A different outcome measure might show something else. Different patients might respond better or worse. The finding is important but preliminary.

Now imagine a year passes, and two other independent research teams run similar trials pairing a heart-rate drug with coordinated non-medication care in similar patients. Both get the same pattern: the drug alone does not move symptoms, but drug-plus-coordinated-care does. Now you have replication — three independent teams, three trials, same pattern. At that point, the finding has graduated from preliminary to fairly robust.

When you read a study describing a treatment or mechanism, ask: *Is this from one team or have others confirmed it?* If it is one team with one study, that finding is worth knowing about, but it is preliminary. If multiple independent teams have confirmed it, even with different methods or patient populations, you can have more confidence it reflects something real.

6.6 Contested findings: knowing when experts disagree

Sometimes you will read about a proposed mechanism for infection-associated illness that sounds compelling, and then you will read a paper that says the opposite, or at least raises major questions. When this happens, the most important skill is recognizing that *this is a real disagreement, not an error by one team*.

A real example: microclots. Some researchers propose that long COVID involves abnormal fibrin clots in the blood that are resistant to the body's normal clot-breaking machinery (Kell, Laubscher, and Pretorius 2022). This would be an important mechanism. But other researchers argue that (Hunt et al. 2024):

- Microclots are not unique to long COVID: similar particles show up in healthy people, in acute COVID-19, and in diabetes
- The correlation between microclot presence and symptoms has not been established
- A trial of clot-breaking therapy has not been done to test whether clearing clots improves symptoms

This is not "one side is right and one is wrong." This is a genuine scientific dispute where both sides have raised legitimate points. The first group says "we have found this abnormality and it appears associated with long COVID." The sec-

ond group says "yes, but the abnormality is not specific and the causal link is not proven." Both statements can be true at the same time.

When you read about a contested mechanism, the honest thing to do is *hold both perspectives*. The mechanism might be real. The evidence might be strong enough to be worth investigating. But it is not yet proven, and the limitations the critics raise are real.

This is where the research state actually lives: most mechanisms under study for infection-associated illness are contested or partially supported, not fully proven. That is not a weakness in the research. That is the normal state of active science. You are reading the *frontier* of what we understand, not a settled textbook of established facts.

6.7 Reading a headline and checking the paper

Let's walk through a concrete example of how to apply all of this.

You read a headline: "New Study Finds Common Virus Causes Long COVID Symptoms." You want to know whether to believe it.

Step 1: Find the actual paper or a reliable summary of it, not just the headline. Headlines are written by journalists trying to be compelling, not by researchers describing their actual findings. The paper itself is more accurate.

Step 2: Check the case definition. Did they study people who met formal long COVID criteria, or people who self-reported symptoms? Did they use WHO, RECOVER, or NASEM definitions? Is the population they studied relevant to you?

Step 3: Look at the study design. How many people? Do they have an uninfected control group? Was it blinded? Single site or multiple sites?

Step 4: Check whether this is a single study or whether other teams have confirmed it. If this is the first report, it is preliminary. If three teams have independently confirmed it, that is much stronger.

Step 5: Look at the confidence intervals and effect sizes. A headline that says "virus causes symptoms" is vague. The actual finding might be "virus was detected in 31% of long COVID patients and 20% of recovered patients, a difference that reaches statistical significance." That is a weaker claim than the headline suggests.

Step 6: Look for alternative explanations. Did the researchers discuss other reasons the finding might show up? Did they test those alternatives?

Step 7: Think about replicability. If the result is real, what would need to be true for other teams to find the same thing? Is the finding specific to the population studied, or would it show up in other populations?

This sounds like a lot of steps, but after reading three or four papers this way, it becomes automatic. You start to see the patterns: what makes a finding robust, what makes it preliminary, what makes it contested.

6.8 What to do with what you know

After you have read the research, you know more than most people about your illness. You know which mechanisms are established and which are hypotheses. You know why different studies report different numbers and how to tell a rigorous study from a preliminary one. You know when experts agree and when they disagree, and what it means in both cases.

The next step is the hardest: deciding what this knowledge means *for you*. A study showing that a mechanism is plausible is not the same as a treatment you can start today. Evidence that something might help is not evidence that it will help you specifically. This is where the scientist's humility and your own clinical judgment have to work together.

If you are talking with a clinician about your care, you can now ask the right questions about any study they cite: *Which definition did they use? How many people? Do they have controls? Has anybody else confirmed this?* A clinician who can answer these questions clearly is one you can trust. One who gets defensive or vague might be repeating something they half-remember, not something they have actually verified.

If you are reading research on your own to understand your illness better, you now know how to distinguish a genuine advance from a preliminary observation, and a contested hypothesis from an established fact. That knowledge will not give you certainty — the research is still evolving and the answers are still unfolding — but it will give you a much more accurate map of the territory.

The map is not the cure. But an accurate map of what is actually known, and what is still unknown, is how you navigate your own illness with clear eyes.

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